

# SymPy Bug Card

## Bug 34: Branch Cut Error in Radical Solving

**Task.** Solve the principal-square-root equation over the complex domain:

$$\sqrt{1/x} = \frac{1}{\sqrt{x}}.$$

**SymPy's answer.**

$$\text{solveset}\left(\sqrt{1/x} = 1/\sqrt{x}, x, \mathbb{C}\right) = \mathbb{C} \setminus \{0\}.$$

**Correct answer.**

$$\mathbb{C} \setminus (-\infty, 0].$$

**Explanation.** Substituting  $x = -1$  gives  $\sqrt{1/(-1)} = i$ , but  $1/\sqrt{-1} = 1/i = -i$ . Thus the residual is  $2i$ , so the negative real values included by SymPy are not solutions.

**Likely root cause.** The diagnosis traces the result through `_solve_radical` in `sympy/solvers/solveset.py`: radical removal turns the checked equation into a broad infinite complement, and that complement is not rechecked against the original branch-sensitive equation. This is a new instance of a known failure mode.

**Artifact (GitHub).** [bug-report/bug-034-branch-cut-error-in-radical-solving/](#)